
Lecture 12: Naïve Bayes Classification

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Bayes Theorem

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

A, B = events

$P(A|B)$ = probability of A given B is true

$P(B|A)$ = probability of B given A is true

$P(A), P(B)$ = the independent probabilities of A and B

- Derived from conditional probability: likelihood of an event or outcome occurring, based on the occurrence of a previous event or outcome



Naïve Bayes: innocent/unsophisticated

- Dataset: $[X, y] = [X_1, X_2, X_3, \dots, X_n, y]$
- Here features: $X_1, X_2, X_3, \dots, X_n$
- The name Naïve Biased is used because it assumes the features go into the model are independent of each other
- Changing the value of one of the features, does not directly influence or does not change the value of other features in this algorithm



Naïve Bayes for multi Variables

- When there are multiple X variables, we assume that they are independent from one another

- Bayes Rule:
$$P(Y = k|X) = \frac{P(X|Y = k) * P(Y = k)}{P(X)}$$

- Here, k is class of Y

- So, the Naïve Bayes for multiple X variables:

$$P(Y = k|X) = \frac{P(X_1|Y = k) * P(X_2|Y = k) \dots P(X_n|Y = k) * P(Y = k)}{P(X_1) * P(X_2) * \dots * P(X_n)}$$



Naïve Bayes for multi Variable

- Here. $P(X_1) * P(X_2) * \dots * P(X_n)$ Is constant
- So we can say,

$$P(Y = k|X) \approx P(X_1|Y = k) * P(X_2|Y = k) \dots P(X_n|Y = k) * P(Y = k)$$



Example of the Naïve Bayes Classifier

- Suppose, a dataset is given. Considering the Weather outlook and temperature, a person would like to go outside and play or not.



Weather	Temperature	Play
Sunny	Hot	Yes
Sunny	Mild	Yes
Sunny	Hot	No
Sunny	Mild	No
Rainy	Mild	Yes
Overcast	Hot	Yes
Sunny	Cool	No
Rainy	Mild	Yes
Rainy	Hot	No
Rainy	Cool	Yes
Overcast	Mild	Yes
Rainy	Mild	No
Overcast	Cool	Yes
Overcast	Cool	Yes



Example of the Naïve Bayes Classifier

	Yes	No	P(Y)	P(N)
Sunny	2	3	$P(S Y)$ 2/9	$P(S N)$ 3/5
Overcast	4	0	$P(O Y)$ 4/9	$P(O N)$ 0/5
Rainy	3	2	$P(R Y)$ 3/9	$P(R N)$ 2/5
Total	9	5		

Outlook feature likelihood table

	Yes	No	P(Y)	P(N)
Hot	2	2	$P(H Y)$ 2/9	$P(H N)$ 2/5
Mild	4	2	$P(M Y)$ 4/9	$P(M N)$ 2/5
Cool	3	1	$P(C Y)$ 3/9	$P(C N)$ 1/5
Total	9	5		

Temperature feature likelihood table



Example of the Naïve Bayes Classifier

- Suppose, **Test data: Sunny, Hot**
- Calculate Conditional Probability:

$$P(Y = k|X) \approx P(X_1|Y = k) * P(X_2|Y = k) \dots P(X_n|Y = k) * P(Y = k)$$

- $P(\text{Yes}|\text{Today})$ = Probability of Yes given {Sunny, Hot}=
 $P(S|Y) \times P(H|Y) \times P(Y) = 2/9 \times 2/9 \times 9/14 = 2/63 = 0.031$
- $P(\text{No}|\text{Today})$ = Probability of No given {Sunny, Hot}=
 $P(S|N) \times P(H|N) \times P(N) = 3/5 \times 2/5 \times 5/14 = 3/35 = 0.085$



Example of the Naïve Bayes Classifier

- Normalized: Probability of Yes given {Sunny, Hot}= $0.031 / (0.031 + 0.085) = 0.267$
- Normalized: Probability of No given {Sunny, Hot}= $0.085 / (0.031 + 0.085) = 0.733$
- $P(\text{Yes} | \text{Today}) < P(\text{No} | \text{Today})$
- Therefore, the prediction is No



Zero-Frequency Problem of the Naïve Bayes Classifier

- One of the disadvantages of Naïve-Bayes is that if there is no occurrences of a class label and a certain feature value together then the frequency-based probability estimate will be zero.
- And this will get a zero when all the probabilities are multiplied.
- If probability of Hot given yes would be 0 or $P(H|Y)=0$,
 $P(\text{Yes}|\text{Today}) = \text{Probability of Yes given \{Sunny, Hot\}} =$
 $P(S|Y) \times P(H|Y) \times P(Y) = 2/9 \times 0 \times 9/14 = 0$
- An approach to overcome this 'zero-frequency problem' is to **add one** to the count for every attribute value-class combination when an attribute value doesn't occur with every class value



Different types of Naïve Bayes

- Multinomial Naive Bayes: Most common one. Suitable for **discrete/categorical** data
- Gaussian Naive Bayes: Suitable for **continuous data** where probabilities can be modeled using a Gaussian/Normal distribution
- Bernoulli Naive Bayes: Like MultinomialNB, this classifier is suitable for discrete data. The difference is that while MultinomialNB works with **occurrence counts**, BernoulliNB is designed for **binary/Boolean features**



Example of Gaussian Naive Bayes

Sepal length	Sepal width	Petal length	Petal width	Class
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
⋮	⋮	⋮	⋮	⋮
5.9	3.0	5.1	1.8	virginica

- The iris dataset contains three classes of flowers of 50 samples in each classes, **Versicolor**, **Setosa**, **Virginica**
- Each class contains 4 features: ‘Sepal length’, ‘Sepal width’, ‘Petal length’, ‘Petal width’.



Gaussian Naive Bayes

- $P(\text{Setosa}) = P(\text{Versicolor}) = P(\text{Virginica}) = 50/150 = 1/3$
- Calculate Mean and Standard Deviation for all 3 classes:
 1. (a) **Mean_SL_Setosa, Mean_SW_Setosa, Mean_PL_Setosa, Mean_PW_Setosa**
(b) **STD_SL_Setosa, STD_SW_Setosa, STD_PL_Setosa, STD_PW_Setosa**
 2. (a) **Mean_SL_Versicolor, Mean_SW_Versicolor,, Mean_PL_Versicolor,, Mean_PW_Versicolor,**
(b) **STD_SL_Versicolor, STD_SW_Versicolor,, STD_PL_Versicolor,, STD_PW_Versicolor,**
 3. (a) **Mean_SL_Virginica, Mean_SW_Virginica, Mean_PL_Virginica, Mean_PW_Virginica**
(b) **STD_SL_Virginica, STD_SW_Virginica, STD_PL_Virginica, STD_PW_Virginica**



Gaussian Naive Bayes

- modeled with Gaussian distribution

$$P(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp \left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2} \right)$$

- Test sample:

SL = 4.85, SW = 3.15, PL = 1.36, PW = 1.12



Gaussian Naive Bayes

- $P_1 = P(\text{Setosa} | TS) = P(TS | \text{Setosa}) \times P(\text{Setosa}) = P(SL | \text{Setosa}) \times P(SW | \text{Setosa}) \times P(PL | \text{Setosa}) \times P(PW | \text{Setosa}) \times P(\text{Setosa})$
- $P_2 = P(\text{Versicolor} | TS) = P(TS | \text{Versicolor}) \times P(\text{Versicolor}) = P(SL | \text{Versicolor}) \times P(SW | \text{Versicolor}) \times P(PL | \text{Versicolor}) \times P(PW | \text{Versicolor}) \times P(\text{Versicolor})$
- $P_3 = P(\text{Virginica} | TS) = P(TS | \text{Virginica}) \times P(\text{Virginica}) = P(SL | \text{Virginica}) \times P(SW | \text{Virginica}) \times P(PL | \text{Virginica}) \times P(PW | \text{Virginica}) \times P(\text{Virginica})$
- Compare P_1 , P_2 and P_3



Drawback

- When does not the conditional independence assumption work?
- It doesn't work in some domains, such as image classification
- Consider a handwritten digit image recognition problem
- Say that we model an image as a vector of pixel values
- But single pixels are not independently generated, depending only on the label



Drawback

- For example, if we look at a black pixel (grayscale value 0) at one of the corners of the image, the pixels around are also very likely to be black
- Thus, the inductive assumption about the conditional independence will not hold in these scenarios

